Midterm Topic Overview

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The midterm exam will take place on Monday October 24, in person for regular students and synchronously on Zoom for most CVN and hybrid students. The make-up date is October 25th.

The exam is closed notes, closed books, no electronic devices. You should have a calculator.

(*) indicates that you should be familiar with the basic concept, but details will not be tested on the exam.

General Linguistics Concepts (Parts of J&M 2nd ed. Ch 1 but mostly split over different chapters in J&M. Ch.1 not yet available in 3rd ed.)

- Levels of linguistic representation: phonetics/phonology, morphology, syntax, semantics, pragmatics
- Ambiguity, know some examples in syntax and semantics, including PP attachment, noun-noun compounds.
- Garden-path sentences.
- Type/Token distinction.
- Know the following terms: word form, stem, lemma.
- Parts of speech:
  - know the 9 traditional POS and some of the Penn Treebank tags (but no need to remember the complete tagset).
- Types of Linguistic Theories (Prescriptive, Descriptive, Explanatory)
- Syntax:
  - Constituency and Recursion. Constituency tests (*).
  - Dependency.
  - Grammatical Relations.
  - Subcategorization
  - Long-distance dependencies. (*)
  - Syntactic heads (connection between dependency and constituency structure).
  - Center embeddings.
  - Dependency syntax:
    - Head, dependent
    - Dependency relations and labels
    - Projectivity

Text Processing (Split over different chapters in J&M. Parts of J&M 3rd ed. Ch. 6

- Tokenization (word segmentation).
- Sentence splitting.
- Lemmatization.

Probability Background

- Prior vs. conditional probability.
- Sample space, basic outcomes
- Probability distribution
- Events
- Random variables
- Bayes' rule
- conditional independence
- discriminative vs. generative models
- Noisy channel model.
- Calculating with probabilities in log space.

Text Classification (J&M 3rd ed. Ch. 6

- Task definition and applications
- Document representation: Set/Bag-of-words, vector space model
- Naive Bayes' and independence assumptions in text classification.

Language Models (J&M 2nd ed. Ch 4.1-4.8, J&M 3rd ed. Ch 4)

- Task definition and applications
- Probability of the next word and probability of a sentence
- Markov independence assumption.
- n-gram language models.
- Role of the START and END markers.
- Estimating ngram probabilities from a corpus:
  - Maximum Likelihood Estimates
  - Dealing with unseen Tokens
  - Smoothing and Back-off:
    - Additive Smoothing
    - Discounting
    - Linear Interpolation
    - Katz' Backoff
- Perplexity

Sequence Labeling (POS tagging) (J&M 2nd ed Ch 5.1-5.5, J&M 3rd ed. Ch 10.1-10.4

- Linguistic tests for part of speech (*).
- Hidden Markov Model:
  - Observations (sequence of tokens)
  - Hidden states (sequence of part of speech tags)
  - Transition probabilities, emission probabilities
  - Markov chain
  - Three tasks on HMMs: Decoding, Evaluation, Training
    - Decoding: Find the most likely sequence of tags:
      - Viterbi algorithm (dynamic programming, know the algorithm and data structures involved)
    - Evaluation: Find the probability of a sequence of words
      - Spurious ambiguity: multiple hidden sequences lead to the same observation.
      - Forward algorithm (difference to Viterbi).
    - Training: We only discussed maximum likelihood estimates.
  - Extending HMMs to trigrams. (*)
- Applying HMMs to other sequence labeling tasks, for example Named Entity Recognition
  - B.I.O. tags for NER.

Parsing with Context Free Grammars (J&M 2nd ed Ch. 12 and 13.1-13.4 and 14.1-14.4 and Ch. 16,
J&M 3rd ed. Ch. 11.1-11.5
and Ch 13.1-13.4

- Tree representation of constituency structure. Phrase labels.
- CFG definition: terminals, nonterminals, start symbol, productions/rules
- derivations and language of a CFG.
- derivation trees (vs. derived string), parse tree.
- Regular grammars / languages(*)
- Complexity classes.
  - "Chomsky hierarchy"
  - Center embeddings as an example of a non-regular phenomenon.
  - Cross-serial dependencies as an example of a non-context-free phenomenon. (*)
- Probabilitistic context free grammars (PCFG):
  - Probability of a tree vs. probability of a sentence.
  - Maximum likelihood estimates from treebanks.
  - Computing the most likely parse tree using CKY.
- Recognition (membership checking) problem vs. parsing
- Top-down vs. **bottom-up parsing.**
- **CKY parser:**
  - bottom-up approach.
  - Chomsky normal form.
  - Dynamic programming algorithm. (know the algorithm and required data structure: CKY parse table). Split position.
  - Backpointers.
  - Parsing with PCFGs

Dependency parsing (J&M 3rd ed. Ch 14.1-14.5
Supplementary material: Küber, McDonald, and Nivre (2009): Dependency Parsing, Ch.1 to 4.2)

- Transition based dependency parsing:
  - States (configuration): Stack, Buffer, Partial dependency tree A
  - Transitions (Arc-standard system): Shift, Left-Arc, Right-Arc
  - Predicting the next transition using discriminative classifiers.
    - Feature definition (address + attribute)
  - Training the parser from a treebank:
    - Oracle transitions from annotated dependency tree.
  - Arc-eager system (*)
- Graph-based dependency parsing (* know the general idea: each word needs to get a head, start with completely connected graph, score each edge, keep the highest scoring edges)

Machine Learning (Some textbook references below)

- Generative vs. discriminative algorithms
- Supervised learning. Classification vs. regression problems, multi-class classification
- Loss functions: Least squares error (*). Classification error.
- Training (empirical risk) vs. testing error (validation risk). Overfitting.
- Linear Models.
  - activation function. (step function)
  - Bias weight
  - perceptron learning algorithm (*)
  - linear separability and the XOR problem. (*)
- Feature functions